| NAME | M. Asad Imtiaz |
|---|---|
| Roll No. | FA24-BCS-062 |
| Section | B |
| Course | Web technologies |
| Instructor | Sir Usman Akram |
| Lab Assignment No. | 2 |

# Lab 2 — Responsive Navbar and Grid Drill

**Difficulty:** Easy–Medium · **Covers:** Lectures 7–10 · **Folder:** `lab-2/`

A focused, smaller-scale rehearsal for Assignment 2: build **just** a responsive navbar (logo left, links center, cart icon right, collapsing into a hamburger menu below 768px) and a 4-column-desktop / 2-column-tablet / 1-column-phone product grid, using Flexbox or Grid plus Bootstrap or Tailwind. Content can be placeholder text/images — the point is the responsive mechanics, not the final design.

**Deliverables in `lab-2/`:** `index.html`, your CSS/framework files, and your completion PDF (with screenshots at phone/tablet/desktop widths).

## Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Responsive Product Page</title>

    <!-- Bootstrap CSS -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
          rel="stylesheet">
```

```
    <!-- Bootstrap Icons -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.13.1/font/bootstrap-
icons.min.css"
          rel="stylesheet">

    <style>
        /* Navbar */
        .navbar {
            padding: 16px 5%;
            background-color: white;
            border-bottom: 1px solid #ddd;
        }

        .navbar-brand {
            font-size: 1.5rem;
            font-weight: bold;
        }

        .navbar-collapse {
            flex-grow: 1;
        }

        .navbar-nav {
            display: flex;
            justify-content: center;
            flex-grow: 1;
            gap: 24px;
        }

        .cart-icon {
            font-size: 1.4rem;
            color: black;
            text-decoration: none;
        }

        /* Product cards */
        .product-card {
            padding: 12px;
            border: 1px solid #ddd;
            border-radius: 8px;
            height: 100%;
        }

        .product-card img {
            width: 100%;
            height: 220px;
            object-fit: cover;
            border-radius: 5px;
        }

        .product-card h5 {
            margin-top: 12px;
        }
```

```
        /* Keep navbar links centered on larger screens */
        @media (min-width: 768px) {
            .navbar-collapse {
                display: flex ;
                align-items: center;
            }
        }
    </style>
</head>

<body>

    <!-- Responsive Navbar -->
    <nav class="navbar navbar-expand-md">
        <div class="container-fluid p-0">

            <!-- Logo: Left -->
            <a class="navbar-brand" href="#">SHOP</a>

            <!-- Hamburger: Visible below 768px -->
            <button class="navbar-toggler"
                    type="button"
                    data-bs-toggle="collapse"
                    data-bs-target="#navbarLinks"
                    aria-controls="navbarLinks"
                    aria-expanded="false"
                    aria-label="Toggle navigation">
                <span class="navbar-toggler-icon"></span>
            </button>

            <!-- Collapsible Links -->
            <div class="collapse navbar-collapse" id="navbarLinks">
                <ul class="navbar-nav">
                    <li class="nav-item">
                        <a class="nav-link" href="#">Home</a>
                    </li>
                    <li class="nav-item">
                        <a class="nav-link" href="#">Products</a>
                    </li>
                    <li class="nav-item">
                        <a class="nav-link" href="#">About</a>
                    </li>
                    <li class="nav-item">
                        <a class="nav-link" href="#">Contact</a>
                    </li>
                </ul>
            </div>

            <!-- Cart: Right -->
            <a href="#" class="cart-icon ms-3" aria-label="Shopping cart">
                <i class="bi bi-cart3"></i>
            </a>
```

```
        </div>
    </nav>

    <!-- Product Grid -->
    <main class="container py-4">
        <h2 class="mb-4">Our Products</h2>

        <div class="row row-cols-1 row-cols-md-2 row-cols-lg-4 g-4">

            <div class="col">
                <div class="product-card">
                    <img src="https://placehold.co/400x300?text=Product+1"
                         alt="Product 1">
                    <h5>Product One</h5>
                    <p>Placeholder product description.</p>
                    <strong>$25.00</strong>
                </div>
            </div>

            <div class="col">
                <div class="product-card">
                    <img src="https://placehold.co/400x300?text=Product+2"
                         alt="Product 2">
                    <h5>Product Two</h5>
                    <p>Placeholder product description.</p>
                    <strong>$30.00</strong>
                </div>
            </div>

            <div class="col">
                <div class="product-card">
                    <img src="https://placehold.co/400x300?text=Product+3"
                         alt="Product 3">
                    <h5>Product Three</h5>
                    <p>Placeholder product description.</p>
                    <strong>$40.00</strong>
                </div>
            </div>

            <div class="col">
                <div class="product-card">
                    <img src="https://placehold.co/400x300?text=Product+4"
                         alt="Product 4">
                    <h5>Product Four</h5>
                    <p>Placeholder product description.</p>
                    <strong>$35.00</strong>
                </div>
            </div>

            <div class="col">
                <div class="product-card">
                    <img src="https://placehold.co/400x300?text=Product+5"
                         alt="Product 5">
                    <h5>Product Five</h5>
```

```
                    <p>Placeholder product description.</p>
                    <strong>$45.00</strong>
                </div>
            </div>

            <div class="col">
                <div class="product-card">
                    <img src="https://placehold.co/400x300?text=Product+6"
                        alt="Product 6">
                    <h5>Product Six</h5>
                    <p>Placeholder product description.</p>
                    <strong>$20.00</strong>
                </div>
            </div>

            <div class="col">
                <div class="product-card">
                    <img src="https://placehold.co/400x300?text=Product+7"
                        alt="Product 7">
                    <h5>Product Seven</h5>
                    <p>Placeholder product description.</p>
                    <strong>$50.00</strong>
                </div>
            </div>

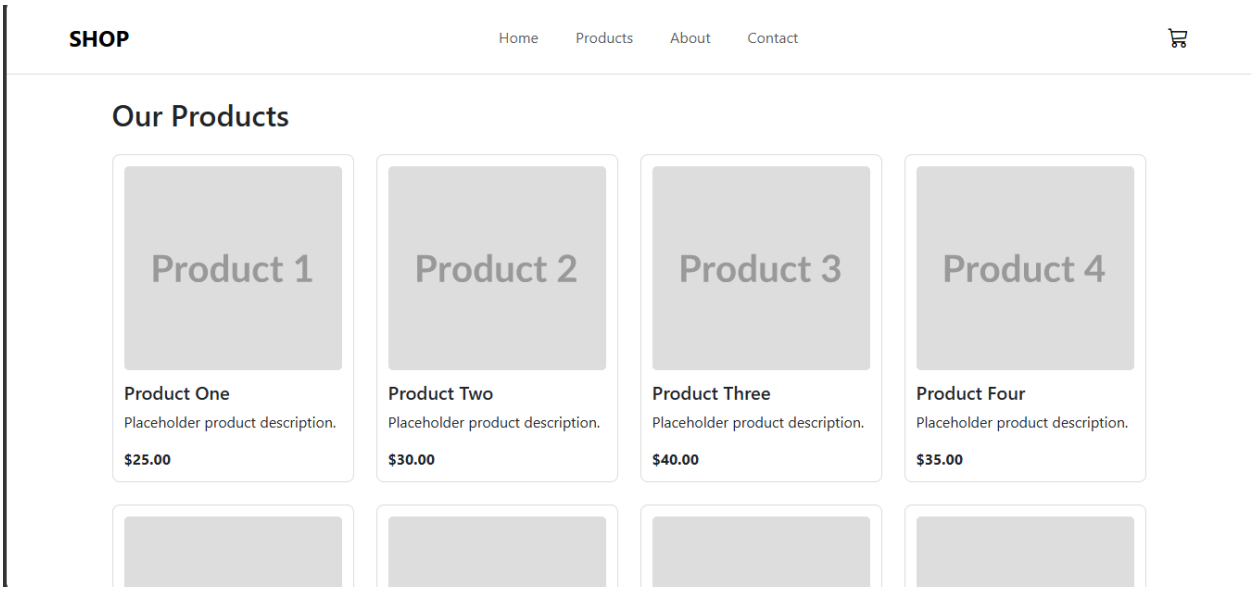
            <div class="col">
                <div class="product-card">
                    <img src="https://placehold.co/400x300?text=Product+8"
                        alt="Product 8">
                    <h5>Product Eight</h5>
                    <p>Placeholder product description.</p>
                    <strong>$28.00</strong>
                </div>
            </div>

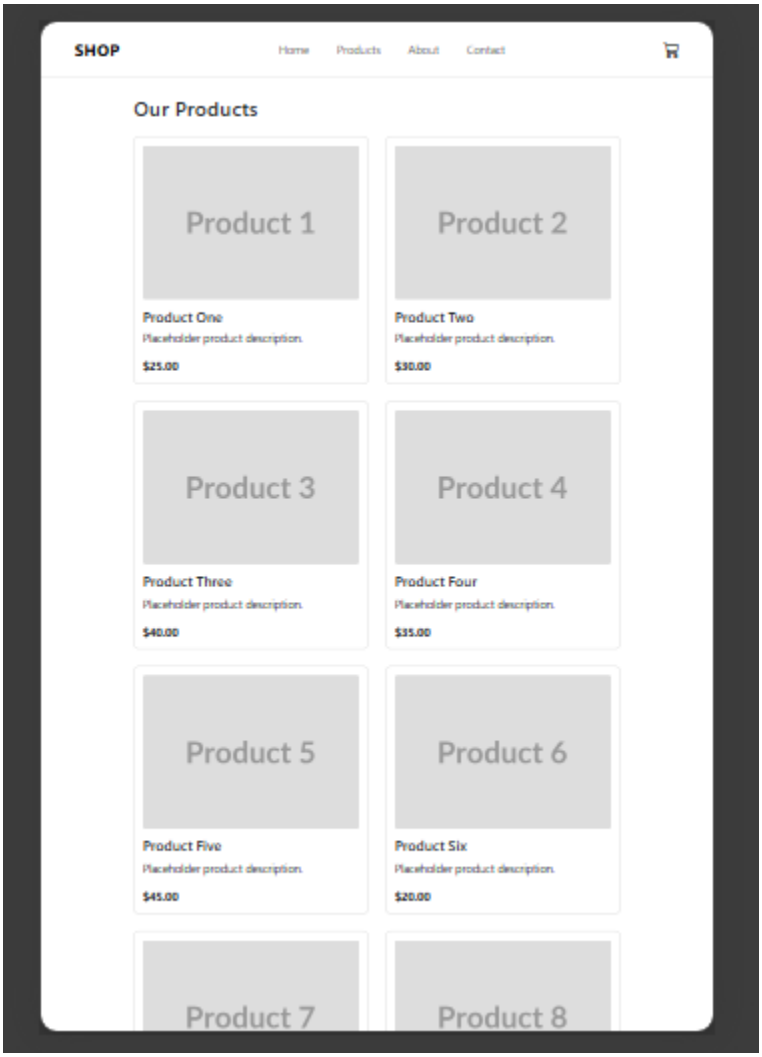
        </div>
    </main>

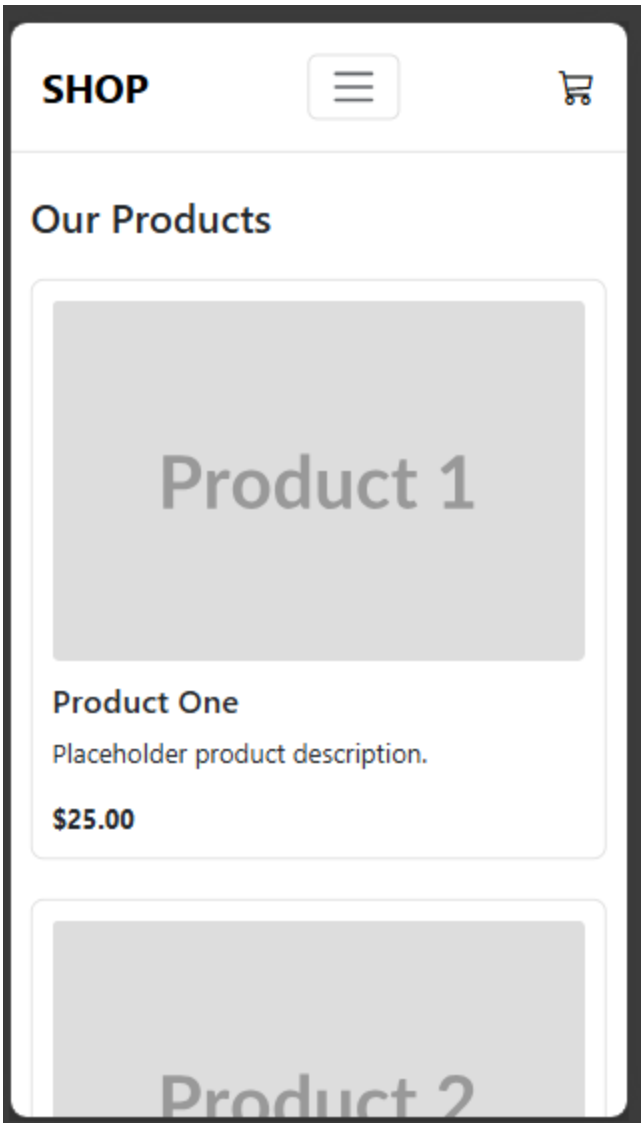
    <!-- Bootstrap JavaScript: enables hamburger toggle -->
    <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"></script>

</body>
</html>
```

## OUTPUT:


SHOP
Home
Products
About
Contact
Our Products
Product 1
Product One
Placeholder product description.
$25.00
Product 2
Product Two
Placeholder product description.
$30.00
Product 3
Product Three
Placeholder product description.
$40.00
Product 4
Product Four
Placeholder product description.
$35.00


SHOP
Home
Products
About
Contact
Our Products
Product 1
Product One
Placeholder product description.
$25.00
Product 2
Product Two
Placeholder product description.
$30.00
Product 3
Product Three
Placeholder product description.
$40.00
Product 4
Product Four
Placeholder product description.
$35.00
Product 5
Product Five
Placeholder product description.
$45.00
Product 6
Product Six
Placeholder product description.
$20.00
Product 7
Product 8


SHOP
Our Products
Product 1
Product One
Placeholder product description.
$25.00
Product 2